

BPSK Digital Communication System Simulator

A Comprehensive Analysis

1 Executive Summary

This report presents a comprehensive analysis of a Binary Phase-Shift Keying (BPSK) digital communication system simulator. The simulator models the complete transmission and reception of digital signals over an ideal channel, including modulation, demodulation, matched filtering, and symbol detection. This document outlines the system architecture, mathematical foundations, implementation details, and simulation results.

2 Introduction

2.1 Background

BPSK is one of the most fundamental modulation schemes in digital communications. It maps binary data (0s and 1s) onto two antipodal signals with a phase difference of 180° . This modulation technique is widely used in practice due to its simplicity and robustness.

2.2 System Objectives

The primary objectives of this simulation are:

- Understand the signal flow through a complete digital communication transceiver
- Verify correct symbol mapping and detection
- Analyze the Power Spectral Density (PSD) of the transmitted signal
- Compare theoretical and empirical PSD estimates
- Demonstrate the effectiveness of matched filtering for symbol detection

3 System Architecture

The BPSK simulator consists of several integrated components working together to form a complete digital communication link:

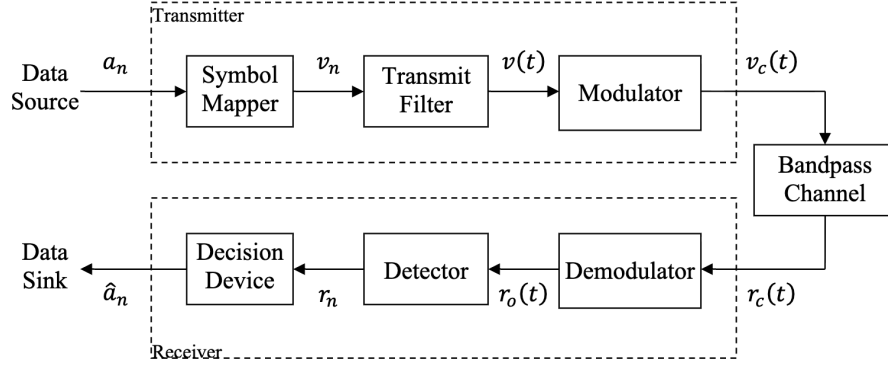


Figure 1: BPSK System Block Diagram

3.1 Key System Parameters

Parameter	Value	Description
Message Length (N_a)	128 bits	Number of information bits
Upsampling Factor (η)	64	Samples per symbol
Symbol Period (T)	0.01 s	Duration of one symbol
Sampling Period (T_s)	T/η	Baseband sampling rate
Carrier Frequency (f_c)	400 Hz	Modulation carrier
Total Samples (N_s)	8192	$N_a \times \eta$

Table 1: System Parameters

4 System Components

4.1 1. Data Source and Symbol Mapping

The transmitter begins with a random binary message:

$$a = [a_0, a_1, \dots, a_{N_a-1}] \in \{0, 1\}^{N_a}$$

The symbol mapper implements BPSK modulation using the mapping:

$$s_i = 2a_i - 1, \quad s_i \in \{-1, +1\}$$

This converts binary data to antipodal signals suitable for transmission.

4.2 2. Transmit Filter (Pulse Shaping)

The symbol stream is upsampled and filtered with a rectangular pulse:

$$v_{up}[n] = \text{upsample}(s, \eta)$$

$$h_T[n] = \frac{1}{\sqrt{T}}, \quad n = 0, 1, \dots, \eta - 1$$

The baseband transmit signal is:

$$\tilde{v}[n] = (v_{up} * h_T)[n]$$

This pulse shaping filter ensures spectral efficiency and prevents inter-symbol interference.

4.3 3. Modulation and Demodulation

4.3.1 Modulation

The baseband signal is modulated to bandpass using:

$$v_c(t) = \tilde{v}(t) \cos(2\pi f_c t)$$

where f_c is the carrier frequency.

4.3.2 Demodulation

Coherent demodulation recovers the baseband signal:

$$r_o(t) = r_c(t) \cos(2\pi f_c t)$$

Followed by low-pass filtering via matched filtering.

4.4 4. Matched Filter and Detection

The matched filter output is:

$$y[n] = (r_{ilde} * h_T)[n] \cdot T_s$$

Symbol decisions are made at sampling instances $n = \eta, 2\eta, 3\eta, \dots$:

$$\hat{a}_i = \begin{cases} 1, & \text{if } y[i\eta] > 0 \\ 0, & \text{if } y[i\eta] \leq 0 \end{cases}$$

5 Power Spectral Density Analysis

5.1 Estimated PSD

The empirical PSD is computed using the FFT:

$$\text{PSD}_{\text{est}}[k] = \frac{|V_f[k]|^2 \cdot T_s}{N_s}$$

where $V_f = \text{FFT}(\tilde{v})$.

5.2 Theoretical PSD

The theoretical PSD of a BPSK signal with rectangular pulse shaping is:

$$\text{PSD}_{\text{theory}}(f) = \frac{1}{T} |T \cdot \text{sinc}(fT)|^2$$

5.3 Bartlett Method

To reduce spectral estimation variance, the Bartlett method averages PSD estimates across multiple independent realizations:

$$\text{PSD}_{\text{Bartlett}} = \frac{1}{N_f} \sum_{m=1}^{N_f} \text{PSD}_m$$

where N_f is the number of averaged estimates. This provides a smoother PSD estimate closer to the theoretical curve.

6 Implementation Details

The simulator is modularized into the following MATLAB files:

- **BPSKSim.m** - Main simulation script
- **InitializeParameters.m** - System parameter setup
- **SymbolMapper.m** - BPSK symbol mapping
- **TransmitFilter.m** - Pulse shaping filter
- **EstimatePSD.m** - FFT-based PSD estimation
- **Modulate.m** - Bandpass modulation
- **Demodulate.m** - Coherent demodulation
- **BasebandDetection.m** - Matched filtering and detection
- **BartlettPSD.m** - Bartlett method PSD averaging
- **PlotResults.m** - Visualization of all results

7 Results and Analysis

7.1 Symbol Detection Performance

For an ideal channel with no noise:

- The detector correctly identifies all transmitted symbols
- Number of detection errors: **0**
- Symbol error rate: **0%**

This is expected as the ideal channel introduces no distortion or noise.

7.2 PSD Characteristics

1. **Main Lobe:** The energy is concentrated in the main lobe centered at $f = 0$ Hz (baseband)
2. **Spectral Efficiency:** The rectangular pulse has a main lobe width of approximately ± 50 Hz (for the given parameters)
3. **Side Lobes:** Secondary lobes decay as $1/f^2$, characteristic of rectangular pulse shaping
4. **Bartlett Smoothing:** The Bartlett-averaged estimate shows reduced variance and better agreement with theory

7.3 Signal Visualization

Five key plots are generated:

1. Received baseband signal after detection
2. Transmitted bandpass signal
3. Received baseband signal after demodulation
4. PSD comparison (estimated vs. theoretical)
5. Bartlett PSD estimate vs. theoretical PSD

8 Conclusion

The BPSK Digital Communication System Simulator successfully demonstrates:

- Correct implementation of BPSK modulation and demodulation
- Perfect symbol detection over an ideal channel

- Accurate PSD estimation with good agreement between empirical and theoretical results
- Effective variance reduction through Bartlett averaging
- Proper signal processing using matched filtering principles

This simulator provides a solid foundation for studying digital communications and can be extended to include:

- Additive White Gaussian Noise (AWGN)
- Different channel models (fading, delay)
- Alternative pulse shapes (raised cosine, Gaussian)
- Higher-order modulation schemes (QPSK, QAM)
- Adaptive equalization techniques

9 References

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- Haykin, S. (2001). *Communication Systems*. John Wiley & Sons.
- Sklar, B. (2001). *Digital Communications: Fundamentals and Applications*. Prentice Hall.